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Midsem Solutions

$$1. \quad X = \begin{cases} 1, & \text{if jam} \\ 0, & \text{if no jam} \end{cases}$$

$$Y | X=0 \sim \text{Exp}(2), \quad Y | X=1 \sim \text{Exp}(1).$$

$$P(X=1) = \frac{1}{3}, \quad P(X=0) = \frac{2}{3}.$$

$$\begin{aligned} P(X=1 | Y=3) &= \frac{f_{Y|X=1}(3) \cdot P(X=1)}{f_{Y|X=1}(3) \cdot P(X=1) + f_{Y|X=0}(3) \cdot P(X=0)} \\ &= \frac{\frac{1}{3} \cdot 1 \cdot e^{-1 \times 3}}{\frac{1}{3} \cdot 1 \cdot e^{-1 \times 3} + \frac{2}{3} \cdot 2 \cdot e^{-2 \times 3}} \\ &= \frac{\frac{1}{3} e^{-3}}{\frac{1}{3} e^{-3} + \frac{4}{3} e^{-6}} = \frac{e^3}{4 + e^3}. \end{aligned}$$

$$2. \quad \underline{X}_1 \sim \text{MVN}_p(0, \Sigma)$$

$$\underline{X}_2 | \underline{X}_1 \sim \text{MVN}_p(s \underline{X}_1, (1-s^2)\Sigma)$$

$$\vdots$$

$$\underline{X}_T | \underline{X}_{T-1} \sim \text{MVN}_p(s \underline{X}_{T-1}, (1-s^2)\Sigma)$$

$$\pi(\underline{X}) = \pi(\underline{X}_1) \pi(\underline{X}_2 | \underline{X}_1) \cdots \pi(\underline{X}_T | \underline{X}_{T-1})$$

$$\begin{aligned} \pi(\underline{X}) &= \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2} \underline{X}_1' \Sigma^{-1} \underline{X}_1} \prod_{t=2}^T \frac{1}{|2\pi(1-s^2)\Sigma|^{1/2}} e^{-\frac{1}{2(1-s^2)} (\underline{X}_t - s \underline{X}_{t-1})' \Sigma^{-1} (\underline{X}_t - s \underline{X}_{t-1})} \\ &= \frac{1}{|2\pi\Sigma|^{T/2} (1-s^2)^{(T-1)P/2}} e^{-\frac{1}{2} \left\{ \underline{X}_1' \Sigma^{-1} \underline{X}_1 + \frac{1}{1-s^2} \sum_{t=2}^T (\underline{X}_t - s \underline{X}_{t-1})' \Sigma^{-1} (\underline{X}_t - s \underline{X}_{t-1}) \right\}} \end{aligned}$$

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$$\pi(\tilde{X}_t | \tilde{X}_1, \dots, \tilde{X}_{t-1}, \tilde{X}_{t+1}, \dots, \tilde{X}_T)$$

$$= \pi(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_{t-1}, \tilde{X}_t, \tilde{X}_{t+1}, \dots, \tilde{X}_T) / \pi(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_{t-1}, \tilde{X}_{t+1}, \dots, \tilde{X}_T)$$

$$\propto \pi(\tilde{X}_1, \tilde{X}_2, \dots, \tilde{X}_{t-1}, \tilde{X}_t, \tilde{X}_{t+1}, \dots, \tilde{X}_T)$$

$$\propto \pi(\tilde{X}_t | \tilde{X}_{t-1}) \pi(\tilde{X}_{t+1} | \tilde{X}_t)$$

$$\propto \frac{1}{|2\pi(1-\rho^2)\Sigma|^{1/2}} \cdot e^{-\frac{1}{2(1-\rho^2)} (\tilde{X}_t - \rho \tilde{X}_{t-1})' \Sigma^{-1} (\tilde{X}_t - \rho \tilde{X}_{t-1})}$$

$$\times \frac{1}{|2\pi(1-\rho^2)\Sigma|^{1/2}} \cdot e^{-\frac{1}{2(1-\rho^2)} (\tilde{X}_{t+1} - \rho \tilde{X}_t)' \Sigma^{-1} (\tilde{X}_{t+1} - \rho \tilde{X}_t)}$$

$$\propto e^{-\frac{1}{2(1-\rho^2)} \left\{ (\tilde{X}_t - \rho \tilde{X}_{t-1})' \Sigma^{-1} (\tilde{X}_t - \rho \tilde{X}_{t-1}) + (\tilde{X}_{t+1} - \rho \tilde{X}_t)' \Sigma^{-1} (\tilde{X}_{t+1} - \rho \tilde{X}_t) \right\}}$$

$$\begin{aligned} & \tilde{X}_t' \Sigma^{-1} \tilde{X}_t (1+\rho^2) - 2\rho \tilde{X}_t' \Sigma^{-1} \tilde{X}_{t-1} - 2\rho \tilde{X}_t' \Sigma^{-1} \tilde{X}_{t+1} + \dots \\ & \quad \downarrow \\ & \text{indep of } \tilde{X}_t \end{aligned}$$

$$\Rightarrow \pi(\tilde{X}_t | \tilde{X}_1, \dots, \tilde{X}_{t-1}, \tilde{X}_{t+1}, \dots, \tilde{X}_T)$$

$$\propto e^{-\frac{1}{2(1-\rho^2)} \left\{ \tilde{X}_t' \Sigma^{-1} \tilde{X}_t \cdot (1+\rho^2) - 2\tilde{X}_t' \Sigma^{-1} (\tilde{X}_{t-1} + \tilde{X}_{t+1}) \rho \right\}}$$

clearly multivariate normal

Suppose  $\tilde{X}_t | \tilde{X}_1, \dots, \tilde{X}_{t-1}, \tilde{X}_{t+1}, \dots, \tilde{X}_T \sim \text{MVN}(A, B)$

$$\pi(\tilde{X}_t | \dots) \propto e^{-\frac{1}{2} (\tilde{X}_t - A)' B^{-1} (\tilde{X}_t - A)}$$

$$= \tilde{X}_t' B^{-1} \tilde{X}_t - 2\tilde{X}_t' B^{-1} A + \dots$$

Comparing the kernels,  $B^{-1} = \frac{1}{1-\rho^2} \Sigma^{-1} (1+\rho^2) \Rightarrow B = \frac{1-\rho^2}{1+\rho^2} \Sigma$

$$B^{-1} A = \Sigma^{-1} \frac{\rho}{1-\rho^2} (\tilde{X}_{t-1} + \tilde{X}_{t+1}) \Rightarrow A = \frac{1-\rho^2}{1+\rho^2} \Sigma \cdot \Sigma^{-1} \frac{\rho}{1-\rho^2} (\tilde{X}_{t-1} + \tilde{X}_{t+1})$$

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$$A = \frac{8}{1+8^2} \cdot (\tilde{X}_{t-1} + \tilde{X}_{t+1})$$

$$\Rightarrow \tilde{X}_t | \tilde{X}_1, \dots, \tilde{X}_{t-1}, \tilde{X}_{t+1}, \dots, \tilde{X}_T \sim \text{MVN} \left( \frac{8}{1+8^2} (\tilde{X}_{t-1} + \tilde{X}_{t+1}), \frac{1-8^2}{1+8^2} \Sigma \right)$$

Q3.  $X_1 | \lambda \sim \text{Exp}(2\lambda)$ ,  $X_t | X_{t-1}, \lambda \sim \text{Exp}(3\lambda X_{t-1})$

$$\lambda \sim \text{Gamma}(a, b)$$

$$\pi(\lambda | X_1, \dots, X_T) \propto \pi(X_1, \dots, X_T | \lambda) \cdot \pi(\lambda)$$

$$\propto \pi(X_1 | \lambda) \pi(X_2 | X_1, \lambda) \dots \pi(X_T | \lambda, X_{T-1}) \cdot \pi(\lambda)$$

$$\propto (2\lambda) \cdot e^{-2\lambda X_1} \cdot \prod_{t=2}^T (3\lambda X_{t-1}) e^{-3\lambda X_{t-1} X_t} \times \lambda^{a-1} e^{-b\lambda}$$

$$\propto \lambda^{T+a-1} e^{-b\lambda - 2\lambda X_1 - 3\lambda \sum_{t=2}^T X_{t-1} X_t}$$

$$\propto \lambda^{T+a-1} e^{-\lambda(b + 2X_1 + 3 \sum_{t=2}^T X_{t-1} X_t)}$$

$$\Rightarrow \lambda | X_1, \dots, X_T \sim \text{Gamma} \left( T+a, b + 2X_1 + 3 \sum_{t=2}^T X_{t-1} X_t \right)$$

Q4.  $\pi(\underline{\theta}) = \pi(\theta_1, \theta_2, \theta_3) = \theta_1^{1-1} \theta_2^{1-1} \theta_3^{1-1} \cdot \frac{\Gamma(1+1+1)}{\Gamma(1)\Gamma(1)\Gamma(1)}$

$$= 2, \quad \theta_1 + \theta_2 + \theta_3 = 1, \quad 0 \leq \theta_1, \theta_2, \theta_3 \leq 1$$

Here,  $\pi(\underline{\theta})$  is the uniform distribution over its support.

$$\underline{Y}_1 = (5, 3, 2)', \quad \underline{Y}_2 = (4, 4, 2)'$$

$$\underline{Y}_3 = (Y_{3,1}, Y_{3,2}, Y_{3,3})', \quad \underline{y}_3 = (y_{3,1}, y_{3,2}, y_{3,3})'$$

$$\pi(Y_{3,1} = y_{3,1}, Y_{3,2} = y_{3,2}, Y_{3,3} = y_{3,3} \mid \underline{Y}_1 = (5, 3, 2)', \underline{Y}_2 = (4, 4, 2)')$$

$$= \pi(\underline{Y}_3 = \underline{y}_3 \mid \underline{Y}_1 = (5, 3, 2)', \underline{Y}_2 = (4, 4, 2)')$$

$$\pi(Y_1 = y_1 \mid \underline{\theta}) = \frac{10!}{5!3!2!} \theta_1^5 \cdot \theta_2^3 \cdot \theta_3^2, \quad \underline{y}_1 = (5, 3, 2)'$$

$$\pi(Y_2 = y_2 \mid \underline{\theta}) = \frac{10!}{4!4!2!} \theta_1^4 \cdot \theta_2^4 \cdot \theta_3^2, \quad \underline{y}_2 = (4, 4, 2)'$$

$$\pi(\underline{Y}_3 = \underline{y}_3 \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2)$$

$$= \int \pi(\underline{Y}_3 = \underline{y}_3 \mid \underline{\theta}) \cdot \pi(\underline{\theta} \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2) d\underline{\theta}$$

$$\pi(\underline{\theta} \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2) \propto \pi(\underline{\theta}) \cdot \pi(\underline{Y}_1 = \underline{y}_1 \mid \underline{\theta}) \cdot \pi(\underline{Y}_2 = \underline{y}_2 \mid \underline{\theta})$$

$$\propto \theta_1^{5+4} \cdot \theta_2^{3+4} \cdot \theta_3^{2+2} = \theta_1^9 \theta_2^7 \theta_3^4$$

$$\propto \theta_1^{10-1} \theta_2^{8-1} \theta_3^{5-1}$$

$$\Rightarrow \underline{\theta} \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2 \sim \text{Dirichlet}(10, 8, 5)$$

$$\Rightarrow \pi(\underline{\theta} \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2) = \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \cdot \theta_1^{10-1} \cdot \theta_2^{8-1} \theta_3^{5-1}$$

$$= \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \theta_1^9 \theta_2^7 \cdot \theta_3^4$$

$$\pi(\underline{Y}_3 = \underline{y}_3 \mid \underline{\theta}) = \frac{(y_{3,1} + y_{3,2} + y_{3,3})!}{y_{3,1}! y_{3,2}! y_{3,3}!} \cdot \theta_1^{y_{3,1}} \cdot \theta_2^{y_{3,2}} \theta_3^{y_{3,3}}$$



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$$\pi(\underline{Y}_3 = \underline{y}_3 \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2)$$

$$= \frac{(y_{3,1} + y_{3,2} + y_{3,3})!}{y_{3,1}! y_{3,2}! y_{3,3}!} \cdot \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \cdot \int \theta_1^{y_{3,1}+9+1-1} \cdot \theta_2^{y_{3,2}+7+1-1} \cdot \theta_3^{y_{3,3}+4+1-1} d\theta_1 d\theta_2 d\theta_3$$

$$= \frac{\Gamma(y_{3,1} + y_{3,2} + y_{3,3} + 1)}{\Gamma(y_{3,1} + 1)\Gamma(y_{3,2} + 1)\Gamma(y_{3,3} + 1)} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(y_{3,1} + 10)\Gamma(y_{3,2} + 8)\Gamma(y_{3,3} + 5)}{\Gamma(y_{3,1} + y_{3,2} + y_{3,3} + 23)}$$

For  $y_{3,1} = 4, y_{3,2} = 4, y_{3,3} = 2,$

Ans of part 2 :

$$\frac{\Gamma(11)}{\Gamma(5)\Gamma(5)\Gamma(3)} \times \frac{\Gamma(23)}{\Gamma(10)\Gamma(8)\Gamma(5)} \times \frac{\Gamma(14)\Gamma(12)\Gamma(7)}{\Gamma(33)}$$

Part 1 :

$$\pi(Y_{3,1} + Y_{3,3} = 9, Y_{3,2} = 1 \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2)$$

$$= \int \pi(Y_{3,1} + Y_{3,3} = 9, Y_{3,2} = 1 \mid \underline{\theta}) \cdot \pi(\underline{\theta} \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2) d\underline{\theta}$$

$\pi(\underline{\theta} \mid \underline{Y}_1 = \underline{y}_1, \underline{Y}_2 = \underline{y}_2)$  same as before.

$$(Y_{3,1}, Y_{3,2}, Y_{3,3}) \mid \underline{\theta} \sim \text{Multinomial}(10, (\theta_1, \theta_2, \theta_3))$$

$$\Rightarrow Y_{3,1} + Y_{3,3} \mid \underline{\theta} \sim \text{Binomial}(10, \theta_1 + \theta_3)$$

$$\pi(Y_{3,1} + Y_{3,3} = 9, Y_{3,2} = 1 \mid \underline{\theta}) = \frac{10!}{9!1!} (\theta_1 + \theta_3)^9 \cdot \theta_2^1$$

$$= \frac{\Gamma(11)}{\Gamma(10)\Gamma(2)} \cdot (1 - \theta_2)^9 \cdot \theta_2^1$$

$$\underline{\theta} \mid \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2 \sim \text{Dirichlet}(10, 8, 5)$$

$$\Rightarrow (\theta_1 + \theta_3) \mid \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2 \sim \text{Beta}(10+5, 8) \\ = \text{Beta}(15, 8)$$

$$\Rightarrow \theta_2 \mid \underline{y}_1 = \underline{y}_1, \underline{y}_2 = \underline{y}_2 \sim \text{Beta}(8, 15)$$

our quantity

$$= \int \frac{\Gamma(11)}{\Gamma(10)\Gamma(2)} \cdot (1-\theta_2)^9 \theta_2^1 \cdot x \frac{\Gamma(23)}{\Gamma(8)\Gamma(15)} \cdot \theta_2^{8-1} (1-\theta_2)^{15-1} d\theta_2$$

$$= \frac{\Gamma(11)\Gamma(23)}{\Gamma(10)\Gamma(2)\Gamma(8)\Gamma(15)} \times \int \theta_2^{9-1} (1-\theta_2)^{24-1} d\theta_2$$

$$= \frac{\Gamma(11)\Gamma(23)}{\Gamma(10)\Gamma(2)\Gamma(8)\Gamma(15)} \times \frac{\Gamma(9)\Gamma(24)}{\Gamma(33)}$$

5.  $\pi(\theta) = \frac{1}{2} \pi_1(\theta) + \frac{1}{2} \pi_2(\theta)$ .

$\pi_1 : \text{Uniform}(0,1) \Rightarrow E_{\pi_1}(\theta) = \frac{1}{2}, V_{\pi_1}(\theta) = \frac{1}{12}$

$\pi_2 : \text{Beta}(2,2) \Rightarrow E_{\pi_2}(\theta) = \frac{1}{2}, V_{\pi_2}(\theta) = \frac{2 \times 2}{(2+2)^2 \times (2+2+1)}$

$E_{\pi}(\theta) = \int \theta \pi(\theta) d\theta = \frac{4}{4 \times 4 \times 5} = \frac{1}{20}$

$= \int \theta \left\{ \frac{1}{2} \pi_1(\theta) + \frac{1}{2} \pi_2(\theta) \right\} d\theta$

$= \frac{1}{2} \int \theta \pi_1(\theta) d\theta + \frac{1}{2} \int \theta \pi_2(\theta) d\theta$

$= \frac{1}{2} E_{\pi_1}(\theta) + \frac{1}{2} E_{\pi_2}(\theta)$

$= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2}$

$E_{\pi}(\theta^2) = \int \theta^2 \pi(\theta) d\theta$

$= \int \theta^2 \left\{ \frac{1}{2} \pi_1(\theta) + \frac{1}{2} \pi_2(\theta) \right\} d\theta$

$= \frac{1}{2} \int \theta^2 \pi_1(\theta) d\theta + \frac{1}{2} \int \theta^2 \pi_2(\theta) d\theta$

$= \frac{1}{2} E_{\pi_1}(\theta^2) + \frac{1}{2} E_{\pi_2}(\theta^2)$

$= \frac{1}{2} \left\{ (E_{\pi_1}(\theta))^2 + V_{\pi_1}(\theta) \right\} + \frac{1}{2} \left\{ (E_{\pi_2}(\theta))^2 + V_{\pi_2}(\theta) \right\}$

$= \frac{1}{2} \left\{ \frac{1}{4} + \frac{1}{12} \right\} + \frac{1}{2} \left\{ \frac{1}{4} + \frac{1}{20} \right\} = \frac{1}{2} \times \frac{4}{12} + \frac{1}{2} \times \frac{6}{20}$

$= \frac{1}{6} + \frac{3}{20}$

$V_{\pi}(\theta) = E_{\pi}(\theta^2) - E_{\pi}(\theta)^2 = \frac{1}{6} + \frac{3}{20} - \frac{1}{4} = \frac{10+9-15}{60} = \frac{4}{60} = \frac{1}{15}$

6.  $X|\beta \sim \text{IG}(\alpha, \beta)$ .

$$L(\beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot x^{-\alpha-1} e^{-\beta/x}.$$

$$\ell(\beta) = \alpha \log(\beta) - \log \Gamma(\alpha) - (\alpha+1) \log(x) - \frac{\beta}{x}.$$

$$\ell'(\beta) = \frac{\partial \ell(\beta)}{\partial \beta} = \frac{\alpha}{\beta} - \frac{1}{x} \Rightarrow \ell''(\beta) = -\frac{\alpha}{\beta^2}.$$

$$\Rightarrow I(\beta) = E(-\ell''(\beta)) = \alpha/\beta^2 \propto \beta^{-2}.$$

$$\Rightarrow \pi(\beta) \propto \sqrt{I(\beta)} \Rightarrow \pi(\beta) \propto \beta^{-1}, \beta > 0.$$

7.  $\pi(\beta | x_1, \dots, x_n) \propto \pi(x_1, \dots, x_n | \beta) \pi(\beta).$

$$\propto \left\{ \prod_{i=1}^n \pi(x_i | \beta) \right\} \times \pi(\beta)$$

$$\propto \left\{ \prod_{i=1}^n \frac{\beta^\alpha}{\Gamma(\alpha)} \cdot x_i^{-\alpha-1} \cdot e^{-\beta/x_i} \right\} \times \beta^{-1}$$

$$\propto \beta^{n\alpha-1} \cdot e^{-\beta \sum_{i=1}^n 1/x_i}$$

$$\Rightarrow \beta | x_1, \dots, x_n \sim \text{Gamma}(n\alpha, \sum_{i=1}^n 1/x_i).$$

8.  $\log \pi(\beta | x_1, \dots, x_n) = c + (n\alpha-1) \log(\beta) - \beta \sum_{i=1}^n \frac{1}{x_i}$

$$\Rightarrow \frac{\partial}{\partial \beta} \log \pi(\beta | x_1, \dots, x_n) = (n\alpha-1) \cdot \frac{1}{\beta} - \sum_{i=1}^n \frac{1}{x_i}$$

$$\frac{\partial^2}{\partial \beta^2} \log \pi(\beta | x_1, \dots, x_n) = -(n\alpha-1) \cdot \frac{1}{\beta^2} < 0.$$

$$\Rightarrow \text{Solve } \frac{\partial}{\partial \beta} \log \pi(\beta | x_1, \dots, x_n) = 0 \Rightarrow (n\alpha-1) \frac{1}{\hat{\beta}_{\text{MAP}}} = \sum_{i=1}^n \frac{1}{x_i}$$
$$\text{for } \beta = \hat{\beta}_{\text{MAP}} \Rightarrow \hat{\beta}_{\text{MAP}} = \frac{n\alpha-1}{\sum_{i=1}^n 1/x_i}.$$



$$E(\hat{\beta}_{\text{MAP}}) = E\left(\frac{n\alpha-1}{\sum_{i=1}^n 1/x_i}\right) = (n\alpha-1) E\left(\left[\sum_{i=1}^n 1/x_i\right]^{-1}\right)$$

Now,  $X_i \stackrel{\text{iid}}{\sim} \text{IG}(\alpha, \beta) \Rightarrow 1/x_i \stackrel{\text{iid}}{\sim} \text{Gamma}(\alpha, \beta)$ .

$$\Rightarrow \sum_{i=1}^n \frac{1}{x_i} \sim \text{Gamma}(n\alpha, \beta) \Rightarrow \frac{1}{\sum_{i=1}^n \frac{1}{x_i}} \sim \text{IG}(n\alpha, \beta) \Rightarrow E\left(\frac{1}{\sum_{i=1}^n 1/x_i}\right) = \frac{\beta}{n\alpha-1}$$

So,  $E(\hat{\beta}_{\text{MAP}}) = (n\alpha-1) \times \frac{\beta}{n\alpha-1} = \beta$ .

$\Rightarrow \hat{\beta}_{\text{MAP}}$  is unbiased estimator for  $\beta$ .

$$\blacksquare V(\hat{\beta}_{\text{MAP}}) = V\left(\frac{n\alpha-1}{\sum_{i=1}^n 1/x_i}\right) = (n\alpha-1)^2 \cdot V(Y)$$

where  $Y \sim \text{IG}(n\alpha, \beta) \Rightarrow V(Y) = \frac{\beta^2}{(n\alpha-1)^2(n\alpha-2)}$ .

$$\Rightarrow V(\hat{\beta}_{\text{MAP}}) = \beta^2 \times \frac{(n\alpha-1)^2}{(n\alpha-1)^2(n\alpha-2)} = \frac{\beta^2}{n\alpha-2} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Q7.  $X|\theta \sim \text{Bin}(n, \theta)$ .

$$L(\theta) = \binom{n}{x} \cdot \theta^x (1-\theta)^{n-x}$$

$$\ell(\theta) = c + x \log(\theta) + (n-x) \log(1-\theta)$$

$$\ell'(\theta) = \frac{x}{\theta} - \frac{n-x}{1-\theta}$$

$$\ell''(\theta) = -\frac{x}{\theta^2} - \frac{n-x}{(1-\theta)^2}$$

$$\begin{aligned} E(-\ell''(\theta)) &= E\left(\frac{x}{\theta^2} + \frac{n-x}{(1-\theta)^2}\right) \\ &= \frac{1}{\theta^2} \cdot n\theta + \frac{1}{(1-\theta)^2} \cdot n(1-\theta) \\ &= \frac{n}{\theta(1-\theta)} \end{aligned}$$

$$\Rightarrow \pi(\theta) \propto \sqrt{I(\theta)}$$

$$\begin{aligned} &\propto \theta^{-1/2} (1-\theta)^{-1/2} \\ &= \theta^{1/2-1} (1-\theta)^{1/2-1} \end{aligned}$$

$$\Rightarrow \theta \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

let's denote it by

$$\pi_1(\theta) \equiv \text{density of Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

suppose,  $\phi = \theta^3$ .

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$$\Rightarrow X|\phi \sim \text{Bin}(n, \phi^{1/3})$$

$$L(\phi) = \binom{n}{x} (\phi^{1/3})^x (1-\phi^{1/3})^{n-x}$$

$$\ell(\phi) = c + x \log(\phi^{1/3}) + (n-x) \log(1-\phi^{1/3})$$

$$\ell'(\phi) = x \cdot \frac{1}{\phi^{1/3}} \cdot \frac{1}{3} \cdot \phi^{-2/3} + (n-x) \cdot \frac{1}{1-\phi^{1/3}} \cdot -\frac{1}{3} \cdot \phi^{-2/3}$$

$$= \frac{1}{3} \cdot x \cdot \frac{1}{\phi} - \frac{1}{3} x(n-x) \times \frac{1}{\phi^{2/3}(1-\phi^{1/3})}$$

$$\ell''(\phi) = -\frac{1}{3} \cdot x \cdot \frac{1}{\phi^2} - \frac{1}{3} \cdot (n-x) \times \frac{d}{d\phi} [\phi^{-2/3} (1-\phi^{1/3})^{-1}]$$

$$\frac{d}{d\phi} \left( \frac{1}{\phi^{2/3} - \phi} \right) = -\frac{1}{(\phi^{2/3} - \phi)^2} \times \left( \frac{2}{3} \phi^{-1/3} - 1 \right)$$

$$\Rightarrow \ell''(\phi) = -\frac{1}{3} \cdot x \cdot \frac{1}{\phi^2} + \frac{1}{3} \cdot (n-x) \cdot \frac{\frac{2}{3} \phi^{-1/3} - 1}{(\phi^{2/3} - \phi)^2}$$

$$\Rightarrow E(-\ell''(\phi)) = \frac{1}{3} \cdot \frac{1}{\phi^2} \cdot E(x) + \frac{1}{3} \cdot \frac{\frac{2}{3} \phi^{-1/3} - 1}{(\phi^{2/3} - \phi)^2} \times E(n-x)$$

$\parallel$   $n\phi^{1/3}$   $\parallel$   $n(1-\phi^{1/3})$

$$= \frac{n}{3} \cdot \phi^{1/3-2} + \frac{1}{3} \cdot n \cdot (1-\phi^{1/3}) \times \frac{\frac{2}{3} \phi^{-1/3} - 1}{\phi^{4/3} (1-\phi^{1/3})^2}$$

$$= \frac{n}{3} \cdot \phi^{1/3-2} + \frac{n}{3} \cdot \frac{\frac{2}{3} \phi^{-1/3} - 1}{\phi^{4/3} (1-\phi^{1/3})}$$

$$= \frac{n}{3} \cdot \phi^{-4/3} (1-\phi^{1/3})^{-1} \left[ \phi^{-2+\frac{1}{3}+\frac{4}{3}} (1-\phi^{1/3}) - \frac{2}{3} \phi^{-1/3} + 1 \right]$$

$$= \frac{n}{3} \times \phi^{-4/3} (1 - \phi^{1/3})^{-1} \left[ \phi^{-1/3} - \cancel{1} + \frac{2}{3} \phi^{-1/3} + \cancel{1} \right]$$

$$= \frac{n}{3} \times \phi^{-4/3} (1 - \phi^{1/3})^{-1} \times \frac{1}{3} \cdot \phi^{-1/3}$$

$$\propto \phi^{-5/3} (1 - \phi^{1/3})^{-1}.$$

$$\Rightarrow \pi_{\phi}(\phi) \propto \phi^{-5/6} (1 - \phi^{1/3})^{-1/2}$$

$$\pi_{\phi}(\phi) d\phi = \pi_{\theta}(\theta) d\theta.$$

$$\Rightarrow \pi_{\phi}(\phi) = \pi_{\theta}(\theta) \frac{d\theta}{d\phi} \quad \theta = \phi^{1/3} \Rightarrow \frac{d\theta}{d\phi} = \frac{1}{3} \phi^{-2/3}$$

$$\Rightarrow \pi_{\phi}(\phi) \propto \theta^{-1/2} (1 - \theta)^{-1/2} \times \phi^{-2/3}$$

$$\propto \phi^{-1/6} (1 - \phi^{1/3})^{-1/2} \times \phi^{-2/3}$$

$$\propto \phi^{-\frac{1+4}{6}} \cdot (1 - \phi^{1/3})^{-1/2}$$

$$\propto \phi^{-5/6} (1 - \phi^{1/3})^{-1/2}$$

$$= \text{Jeffrey's prior for } \phi.$$

$\Rightarrow$  Jeffrey's prior is invariant under transformation.